📄 E-Commerce Customer Purchase Analysis

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Objective

To analyze customer purchasing behavior in an e-commerce platform and generate meaningful insights that help improve marketing strategies, identify high-value customers, and enhance overall sales performance.

Dataset Description

The dataset contains customer transaction records with the following attributes:

- Customer ID
- Customer Name
- City
- Product Purchased
- Quantity
- Price per unit

Each record represents a single purchase made by a customer.

Tools Used

- SQL (for querying and data aggregation)
- Excel (for pivot tables and quick analysis)
- Python (Pandas, Matplotlib for analysis and visualization)

Data Processing

- Loaded dataset into SQL, Excel, and Python environments
- Cleaned the dataset by checking for missing or inconsistent values
- Created a new column:
  Total Purchase = Quantity × Price
- Structured data for efficient analysis

Analysis Performed

1. **Customer Spending Analysis**
   - Calculated total spending of each customer
   - Identified top customers contributing maximum revenue
2. **Product Popularity Analysis**
   - Determined most purchased products based on quantity
   - Helped understand demand trends
3. **City-wise Revenue Analysis**
   - Computed total revenue generated by each city
   - Identified high-performing locations

Visualizations

- **Bar Chart:** Displays product-wise sales to compare popularity
- **Pie Chart:** Shows percentage distribution of revenue across cities

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Key Insights

- A small group of customers contributes a large portion of total revenue
- Certain products (like electronics) dominate overall sales
- Major cities generate significantly higher revenue compared to smaller cities
- Customer purchase behavior varies based on location and product type

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Conclusion

The analysis highlights important patterns in customer purchasing behavior, enabling businesses to make data-driven decisions. These insights can be used to design targeted marketing campaigns, improve customer retention, and optimize product offerings.

Future Scope

- Perform time-based analysis (monthly/seasonal trends)
- Implement customer segmentation using machine learning
- Build recommendation systems for personalized shopping experience
- Integrate real-time dashboards for continuous monitoring